

# Daiyang He

136-0827-0773 | [hedaiyang@stu.scu.edu.cn](mailto:hedaiyang@stu.scu.edu.cn) | <https://github.com/DaiyangHe> | 2005-06-11 | <https://daiyanghe.github.io/homepage>

## Education

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| 2023–Present                                                                                                                                                                                                                   | <b>Sichuan University, SCUPI</b><br>Industrial Engineering<br>B.S. (GPA: 3.85, 8/73) |
| <ul style="list-style-type: none"><li>Research direction: <b>Mechanism-Informed AI for Clinical Insights</b></li><li>Keywords: AI4Science, Deep Learning, CFD, Physics Informed Machine Learning, System Engineering</li></ul> |                                                                                      |

## Research Papers

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| <ul style="list-style-type: none"><li>Y. Qiu*, <b>Daiyang He*</b>, Z. Jiang, S. Cheng, Y. Tai, K. Li, H. Wu. <i>Open-source and fully automated multi-organ CT analysis for non-invasive portal hypertension evaluation</i>. <i>npj Digital Medicine</i>, Under Review, 2026.</li><li>C. Weng*, <b>Daiyang He*</b>, S. Cheng, D. Yuan, T. Wang, W. Zeng, Y. Luo, J. Wang, K. Li, Y. Qiu. <i>Noise-Pressure Constrained Liutex Method for Robust Vortex Identification in 4D Flow MRI of Abdominal Aortic Aneurysms</i>. <i>Computer Methods and Programs in Biomedicine</i>, Major Revision, 2026.</li><li>S. Cheng, Y. Qiu, <b>Daiyang He</b>, M. L. Wang, C. Weng, D. Yuan, K. Li. <i>Helical vortex arising from aberrant right subclavian artery induces occurrence of dissection tears</i>. <i>Physics of Fluids</i>, 2025.</li><li>S. Cheng, L. Liu, Y. Qiu, Y. Diao, <b>Daiyang He</b>, F. Yan, C. Wang, K. Li. <i>Hypoxia promotes blood blister-like aneurysms development: A CFD study</i>. <i>Journal of Biomechanics</i>, 2025.</li></ul> |  |
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## Academic Competitions and Achievements

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| <ul style="list-style-type: none"><li>2025: <b>Undergraduate Training Program (National Level)</b>, “ShadowSift: An Efficient Real-Time Detection System for Low-Quality Deepfake Videos.”</li><li>2025: <b>Third Prize</b>, National Collegiate Mathematical Contest in Modeling</li><li>2024: <b>Third Prize</b>, Southwest Division of the China Undergraduate Physics Tournament (CUPT)</li><li>2024: <b>S Award</b>, American Collegiate Mathematical Contest in Modeling</li><li>2024–2026: <b>Excellent Student, Outstanding Students Leader, Comprehensive Scholarship</b>, Sichuan University</li></ul> |  |
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## Work Experience

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| 2023–Present | <b>Research Assistant, SCUPI BioFluid Lab</b> <ul style="list-style-type: none"><li>Conduct CT/MRI image analysis and CFD simulation.</li><li>Develop the web-based automatic portal hypertension assessment pipeline <b>ALPHA</b> (<a href="http://scupi.alpha.hxmedbot.com:10910/">http://scupi.alpha.hxmedbot.com:10910/</a>).</li><li>Explore physics-informed machine learning modeling for CFD surrogate.</li></ul> |
| 2024–2026    | <b>Student Organizations, SCUPI</b> <ul style="list-style-type: none"><li><b>Leader</b> of Technology and Science Association (TSA)</li><li><b>Teaching Assistant</b> of IE major core courses (Productivity Analysis, Information System, Interdisciplinary Research)</li><li><b>Peer Advisor</b></li></ul>                                                                                                              |

## Competences & Languages

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| <b>Programming</b>          | Python, C/C++, MATLAB, SQL, $\LaTeX$                                                                                |
| <b>Scientific Knowledge</b> | Medical Image Analysis, Machine Learning, Harness Engineering, CFD, Probability & Statistics, Mathematical Analysis |
| <b>Software</b>             | Mimics, 3D Slicer, ANSYS Fluent, ICEM CFD, ParaView, TecPlot                                                        |
| <b>Languages</b>            | English (fluent reading, writing, speaking), TOEFL 97                                                               |